measured lateness; output mode and state bound justified.
Part 2: Streaming Pipeline 19.5 pts Query correctness (7.5 pts): Window definitions and aggregates are correct on reference replay.
Fault tolerance (7.5 pts): Restart demonstration with exact reconciliation and no duplicate window keys.
Metrics & reproducibility (4.5 pts): Metrics log complete; pinned environment; CI green on the CPU baseline.
GenAI Verification 1.5 pts Includes the required GenAI log (GenAI.pdf) or explicitly states AI was not used.
🏠 Return to Course Home
📝
All HW
⬆️
Module
⏫
Phase
Choose a submission type

ChatGPT dijo:

Dijiste:

Texto pegado(8)....
Documento

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[PS] Homework 06 of 12 (Problem Set): Semantic Search, Chunking, & VDB Schema
svg
In Progress
NEXT UP: Submit Assignment
Unlimited Attempts Allowed
Home » Phase 3: Scrub & Embed » Module 05: Scrub (Chunking, Embedding Generation, VDB Schema Design) » PS 6
📝 All Assignments

ℹ️ **CRITICAL SUBMISSION POLICY (Schema Authority)**

CIS 531 submissions using the standard project must adopt the instructor-provided ontology schema unmodified; CIS 731 submissions must derive their own. **For an approved course project**, a 531 submission containing a self-designed schema, or a 731 submission reusing the unmodified instructor schema, fails the register_schema contract check.

GenAI Kit Requirement: You must include your GenAI.pdf log or explicitly state "No GenAI Used."
THEORY & SCHEMA

Semantic Search, Chunking, & VDB Schema

Chunking and index trade-off analysis, retrieval evaluation design, and a provenance-bearing vector-database schema.

📅 **Assigned:**
Fri 16 Oct 2026

⏰ **Due:**
Sun 25 Oct 2026

💯 **Points:**
30.0 pts

📂 **Submit:**
Online → File Uploads (.pdf, max 8 pages) & Website URL (Git Repo URL containing `schema/` dir under tag `ps6-submission`)

Required Readings

Speech and Language Processing (3rd ed. draft)

- **Chapter 11:** Retrieval, dense embeddings, retrieval-augmented generation, and ranked-retrieval evaluation.

📖 Access Resource →

Sentence Transformers Documentation

- **Semantic Search:** Symmetric vs. asymmetric search, bi-encoder retrieval with cross-encoder re-ranking, and approximate nearest-neighbor search.

📘 Access Resource →

Part 1: Chunking, Index, and Evaluation Analysis

(15.0 Points)
Complete the chunking cost model (6.0 pts), index and similarity analysis (4.5 pts), and retrieval evaluation protocol (4.5 pts) as defined in the assignment spec.

Required Analyses

- Compare fixed-window, sentence-boundary, and structure-aware chunking.
- Contrast IVF and HNSW on query complexity, build/memory cost, and recall degradation.

Part 2: Schema and Contracts

(13.5 Points)
Objective: Define the provenance-bearing vector-record schema and write the admit_document and vdb_upsert contracts.

1. Draft JSON Schema (Draft 2020-12) extending the MP 5 record.
2. Write deal contracts implementing EmbeddingCompleteness and NoOrphanEmbeddings.
3. CIS 531: Map fields to instructor ontology. CIS 731: Derive custom ontology and justify granularity.

Deliverable: Include schema derivations and contract code in your final PDF and submit the schema/ directory via Git tag ps6-submission.

Scoring Rubric

Component	Weight	Description
Part 1: Analysis (Chunking, Index, Evaluation)	15.0 pts	Chunking cost model (6.0 pts), Index and similarity analysis (4.5 pts), Retrieval evaluation protocol (4.5 pts).
Part 2: Schema & Contracts	13.5 pts	Record schema and provenance (6.0 pts), Contracts and invariants (4.5 pts), Section-specific ontology deliverable (3.0 pts).
GenAI Verification	1.5 pts	Includes required GenAI log (GenAI.pdf) or explicitly states AI was not used.

do not give the code but the same instructions for mp6 and make as simple as it caan b, tell the files i need to create since it used data sources the proffesor have not provided just yet, i will my own BTS data for testing

ChatGPT dijo:

Dijiste:

is it ok for me to do the report with ai after i share y code?

ChatGPT dijo:

Dijiste:

MP6_Genesis.zip
Archivo

ChatGPT dijo:

PS6_Genesis_Report_Draft.pdf
PDF

Dijiste:

genesisjoseph@ip-10-150-212-81 MP6_Genesis % ls -l schema/contracts.py schema/tests/test_contracts.py
python3 -m pytest schema/tests -q
-rw-r--r--@ 1 genesisjoseph staff 6574 Oct 29 20:45 schema/contracts.py
-rw-r--r--@ 1 genesisjoseph staff 2825 Oct 29 20:40 schema/tests/test_contracts.py
.....
[100%]
===
warnings summary
===
../../../../../../Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages/dateutil/tz/tz.py:37

/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages/dateutil/tz/tz.py:37: DeprecationWarning: datetime.datetime.utcfromtimestamp() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.fromtimestamp(timestamp, datetime.UTC).
  EPOCH = datetime.datetime.utcfromtimestamp(0)

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
8 passed, 1 warning in 0.22s
genesisjoseph@ip-10-150-212-81 MP6_Genesis %

I WAS ABLE TO TEST IT

ChatGPT dijo:

Dijiste:

UPDATE REPORT MAKE IT SIMPLE EASY FOR INSTRUCTOR TO FOLLOW

Procesó durante 23s >

ChatGPT dijo:

PS6_Genesis_Report_Updated.pdf
PDF